EECE 2311

Lab for EECE2310: Introduction to Digital Design and Computer Architecture

SPRING 2026

Lab Assignment #06

Sequential Circuit Design for Railroad Crossing Signal

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Abstract

This project will simulate a railroad crossing signal control system using finite state machine(FSM)

Introduction

This lab simulate a railroad crossing signal control system. To achieve this goal, you will need to design a finite state machine (FSM) that represents the behavior of the system and then implement the sequential circuit for that FSM . You will use push buttons to represent the railroad sensors and LEDs to represent the signal lights. A switch will be utilized to reset the system to an initial state.

Railroad Crossing Signal System Requirements

The railroad crossing signal system that we need to simulate is represented in Figure 1. The system has the following specifications and requirements:

a. The railroad is one way only, from east to west;
b. As shown, SE and SW represent the east and west sensors respectively. A sensor outputs logic 1 only when a train is passing by it;
c. The trains can be of any length relative to the distance between SE and SW as shown;
d. No back-to-back trains are allowed(i.e., no train is allowed to pass by SE if the rear of the train ahead of it has not passed away from SW yet).
e. SE and SW are assumed to work perfectly with no wrong output, due to debouncing for example.

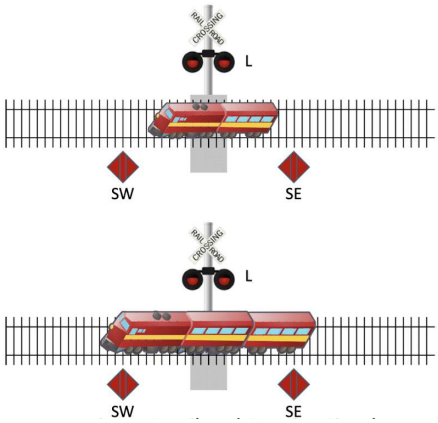


Figure 1: Railroad Crossing Signal

Based on the above requirements, the system can be in any of the following **five states**:

(0) Both SW and SE are off, and the signal light L is also off.

This is the "initial" state of the system, when it is reset.

It is also the "idle" state when there is no train in the crossing area.

☆ While in this state, it is **irregular** for SW to turn on before SE. This is because no trains from the other direction are allowed in this system.

(1) SW is off, SE is on, and hence L is on. This state happens when a train is passing by SE but has not arrived at SW yet.

(2) Both SW and SE are off while L is on. This state happens when a short train is in the area between SE and SW.

☆ It is **irregular** while in this states for SE to turn on. This is because back-to-back trains are

not allowed.

(3) SW is on, SE is off, and L is on. This state happens when a short train arrives at SW.

☆ It is **irregular** while in this state for SE to turn on. This is because back-to-back trains are not allowed.

(4) SW is on, SE is on, and L is on. This state happens when a long train is simultaneously passing by both sensors.

☆ It is **irregular** while in this state for SE to stay on while SW turns off.

The behavior of the system for both states 3 and 4 (in terms of their expected next states) is identical. So, in the finite state diagram, combine these two states in one state. Therefore, the finite state diagram will have only four states, which will save one extra flip-flop.

Explanation:

State(3) and state (4) can be combined because they are behaviorally equivalent in terms of future transitions. They respond identically to every possible inout and produce the same output.

Input(SE,SW)	state(3) next state	state(4) next state
(0, 0)	state(0) L off	state(0) L off
(0, 1)	stay at state(3)	stay at state(4)
(1, 0)	Irregular: stay at state(3)	Irregular: stay at state(4)
(1, 1)	Irregular: stay at state(3)	Irregular: stay at state(4)

Table 1: State (3) and State (4) Equivalence

Pre-Lab Assignment

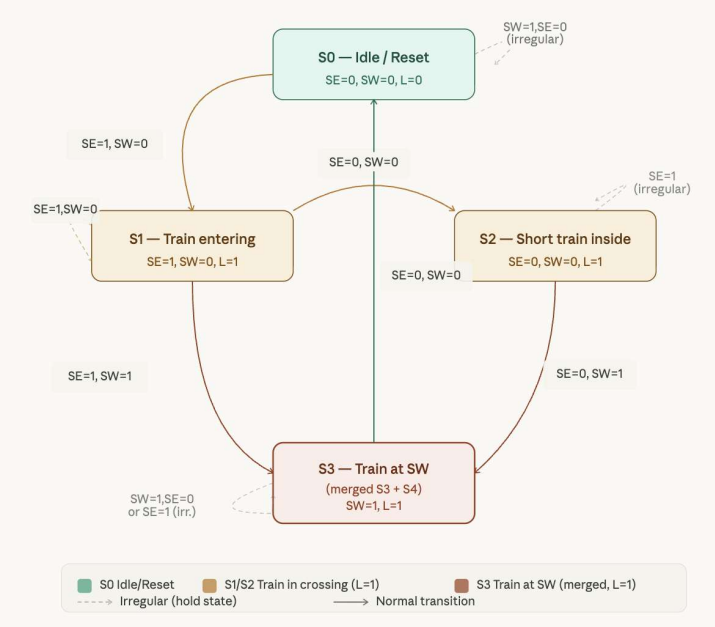


Figure 2: finite state diagram for the system

S0 → S1 (SE=1, SW=0): A train trips the east sensor. The crossing signal L turns on.

S1 → S2 (SE=0, SW=0): The rear of a *short* train clears SE before reaching SW. The train is now between the two sensors; L stays on.

S1 → S3 (SE=1, SW=1): A *long* train simultaneously reaches SW while still covering SE. Goes directly to the merged S3 state; L stays on.

S2 → S3 (SE=0, SW=1): The short train reaches SW. L stays on. This is the "short train at SW" path into the merged state.

S3 → S0 (SE=0, SW=0): The train fully clears SW. Both sensors off, L turns off — system returns to idle.

Irregular transitions (hold state):

- In **S0**: SW=1 before SE=1 — impossible direction, stay in S0.
- In **S2**: SE turns on — would imply a back-to-back train, stay in S2.
- In **S3**: SE turns on — same back-to-back violation, stay in S3.

Sequential Circuit Implementation

State	Q_A	Q_B	Meaning
S0	0	0	Idle: SE = 0, SW =0; L = 0
S1	0	1	Train entering: SE = 1, SW =0, L =1
S2	1	0	Short train inside: SE = 0; SW = 0, L = 1
S3	1	1	Train at SW (merged): SW = 1; L =1

Table 2: State encoding

Current State		Inputs		Next State		Reasoning
QA	QB	SW	SE	DA	DB	
0	0	0	0	0	0	S0 → S0 (idle, stay)
0	0	0	1	0	1	S0 → S1 (train trips SE)
0	0	1	0	0	0	S0 → S0 (irregular SW on first, hold)
0	0	1	1	0	0	S0 → S0 (irregular, hold)
0	1	0	0	1	0	S1 → S2 (train clears SE, short train)
0	1	0	1	0	1	S1 → S1 (train still on SE, stay)
0	1	1	0	1	1	S1 → S1 (irregular SW without SE, hold)
0	1	1	1	1	1	S1 → S3 (long train hits SW)
1	0	0	0	1	0	S2 → S2 (train still between sensors, stay)
1	0	0	1	1	0	S2 → S2 (irregular back-to-back, hold)
1	0	1	0	1	1	S2 → S3 (short train reaches SW)
1	0	1	1	1	0	S2 → S2 (irregular back-to-back, hold)
1	1	0	0	0	0	S3 → S0 (train fully clears SW)
1	1	0	1	1	1	S3 → S3 (irregular SE on while SW off, hold)
1	1	1	0	1	1	S3 → S3 (SW still on, stay)
1	1	1	1	1	1	S3 → S3 (SW still on, stay)

Table 2: State encoding

Boolean expressions:

DA = QA·QB’ + QA’·QB·SE' + QB·SW + QA·SE

DB = QB·SE + QB·SW + QA·SW·SE’ + QA’·SW’·SE

L = QA + QB

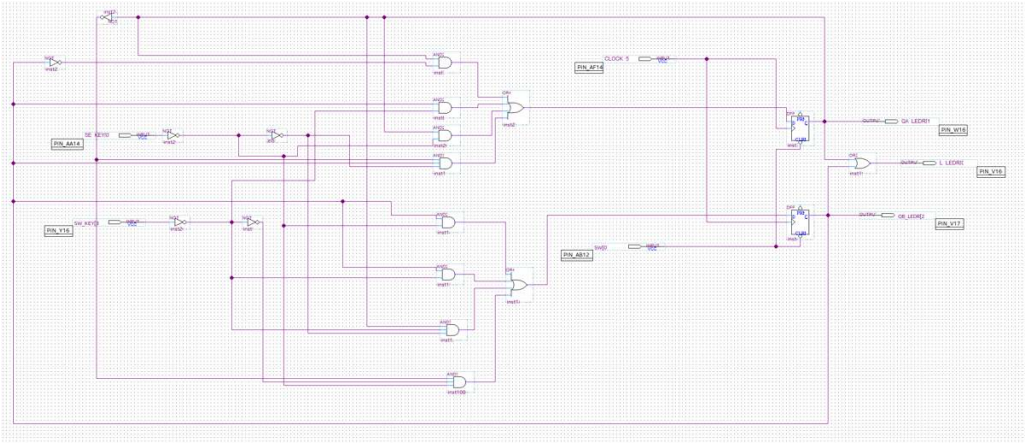


Figure 3: schematic for the system

(short video seen in uploaded files)

Analysis

The railroad crossing signal system was successfully implemented using a four state FSM with two D flip-flops. The state encoding assigned QA and QB as the two state bits, allowing the system to represent all four states.

Conclusion

This lab demonstrated the process of designing and implementing a sequential circuit from a real-world problem description.

References

1. Michael Benjamin, “*Lab Report Guide*”, Northeastern University, January 18, 2006.
2. Given lab 6 pdf in EECE2311 canvas.